

Hui Li

Robotics Research Engineer | Dexterous Manipulation • Sim-to-Real • Real-World Deployment

lihuione@gmail.com | [Demos](#) | [GitHub](#) | [Google Scholar](#) | [LinkedIn](#) | F-1 STEM OPT (valid until 09/2028)

SUMMARY

Robotics Research Engineer specializing in **contact-rich dexterous manipulation systems** that transfer from simulation to hardware and deploy reliably in the real world. Delivered **83% grasp success** across **500+ real-world trials** and **88% sim-to-real transfer** on a **40-DoF bimanual mobile manipulation platform**. Designed **Physically Structured Policy Learning (PSPL)**, improving simulation success by **11 percentage points** and reducing episode length by **30%**. Built **real-time ROS 2 stacks** with **120 Hz closed-loop control** and **sub-ms on-device inference**.

TECHNICAL SKILLS

- **Robotics:** ROS 2 (MoveIt2, Nav2, ros2_control), PyBullet, OpenAI Gym, FoundationPose, URDF
- **Languages & Tools:** Python, PyTorch, OpenCV, Linux, C++, Docker, Git
- **ML/RL:** Reinforcement Learning (PPO, SAC), Deep Learning, Sim-to-Real Transfer, Imitation Learning
- **Sensing:** Visual (RealSense, Zivid), Tactile/FT (Contactile, Robotous, ErgoGlove), SCiO Spectrometer
- **Hardware:** UR5e, Sawyer, Psyonic/Schunk Hands, Husky UGV, Jetson AGX Orin

EXPERIENCE

Lead Robotics Engineer – PhD Research

UA Autonomous Robotic Lab (08/2018–08/2025)

Project | *DexMobile: 40-DoF Bimanual Mobile Manipulation System*

- Built and deployed a **40-DoF bimanual mobile manipulation system**, achieving **83% grasp success** across **500+ real-world trials** under real-time closed-loop control.
- Developed a **sim-to-real pipeline** with a PyBullet digital twin and domain randomization, achieving **88% sim-to-real transfer** in contact-rich tasks.
- Designed **Physically Structured Policy Learning (PSPL)** to structure policy learning around contact-aware physical regimes with physics-grounded transitions, achieving **95% simulation success (+11%)** with **30% shorter episodes**.
- Engineered a **120 Hz real-time control stack** with synchronized 30 Hz perception and **deployed learned policies on Jetson AGX Orin (~0.6 ms inference)** for closed-loop manipulation.
- Authored and open-sourced the **first ROS 2 driver for Psyonic Ability Hand (125 Hz tactile streaming)**, adopted by external labs and recognized by the vendor.
- Analyzed **real-world failure modes** (e.g., slip under limited tactile coverage), motivating integration of **3-axis tactile sensing** for improved contact state estimation.
- **Led system integration across perception, control, and sim-to-real pipelines**; mentored 3 PhD researchers on ROS 2 and real-robot deployment, contributing to **2 co-authored publications**.

Project | *MagicHand: 17-DoF Multi-Modal Manipulation Platform*

- Built a **17-DoF multi-modal manipulation platform** (Sawyer + AR10) with vision/tactile/NIR sensing; collected a **22K+ multi-modal dataset** and **achieved 86% grasp success (+22% over baseline)**.
- Developed an **LLM-based grasp planner (GPT-4)** for feature complementation and task planning, achieving **83% strategy match** and **80% success rate** on real-world objects.

EDUCATION

Ph.D. in Computer Science

University of Alabama (05/2018–08/2025)

- 10 publications, 6 first-authored; venues include ICRA, Humanoids, ICSR

M.S. in Electrical Engineering

Illinois Institute of Technology (08/2008–05/2011)

B.S. in Electrical Engineering

Hebei University of Science and Technology (08/2002–05/2006)

SELECTED PUBLICATIONS

- Physically Structured Policy Learning for Contact-Rich Dexterous Manipulation – **IEEE RA-L** (under review)
- Grasp Intention Interpretation in Object Handover for Human-Robot Teaming – **ICSR 2024** [[PDF](#)]
- Learning Task-Oriented Dexterous Grasping from Human Knowledge – **ICRA 2021** [[PDF](#)]
- Context-Aware Dexterous Grasping Using an Anthropomorphic Robotic Hand – **ICRA 2020** [[PDF](#)]

AWARDS

NSF I-Corps Grant (Entrepreneurial Lead) • Ph.D. Excellence Award • Shocker Innovation Award